

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

20/20
21/21

Code of Conduct:

Q. Santosh Kumar 192514080

Santosh Kumar (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Scenario	20%				
Identification of Key Issues	20%				
Application of Concepts	25%				
Decision Making	20%				
Clarity of Response	15%				

Solution

Base Network : 192.168.100.0/24 we allocate the subnets sequentially to prevent address overlapping.

School of Computing (125)

Network Address : 192.168.100.0

Broadcast Address : 192.168.100.127

Valid Host Range : 192.168.100.1 to 192.168.100.126

Usable Host : $2^7 - 2 = 126$

School of Electronics (126)

Network Address : 192.168.100.128

Broadcast Address : 192.168.100.191

Valid Host Range : 192.168.100.129 to 192.168.100.190

Usable Host : $2^6 - 2 = 62$

School of Mechanical Engineering (127)

Network Address : 192.168.100.192

Broadcast Address : 192.168.100.223

Valid Host Range : 192.168.100.193 to 192.168.100.222

Usable Hosts : $2^5 - 2 = 30$

Reming unused IP Range 192.168.100.224 to 192.168.100.255

destination IP: 192.168.2.45

destination subnet: 192.168.2.0/24

Matching Routing Table Entry: 192.168.2.0/24

Next-hop Network / Interface: the packet will be forwarded out of the specific router interface directly connected to the 192.168.2.0/24 network.

default route: If the destination does not match any entry, the packet is sent to the default Router (commonly represented as 0.0.0.0/0).

Based Network: Admin (192.168.10.0/26)

Finance (192.168.10.64/26)

Administration LAN (192.168.10.0/26)

Network Address: 192.168.10.0

Broadcast Address: 192.168.10.63

Valid Host Ranges: 192.168.10.1 to 192.168.10.62

default Gateway: 192.168.10.1

suggested IPs (Excluding Gateway): 192.168.10.2, 192.168.10.3, 192.168.10.4.

LAN (192.168.10.64/26)

Network Address : 192.168.10.64

Broadcast Address : 192.168.10.127

Valid Host Range : 192.168.10.65 to 192.168.10.126

Default Gateway : 192.168.10.65

Suggested IPs (Excluding Gateway): 192.168.10.66,

192.168.10.67, 192.168.10.68.